

Annex 1

Estimates of the total population, 1800-2050. Sources and procedures

This annex describes the total population estimates for each of the Latin American countries for the period 1800-2050. In each case, the available data are presented first, indicating their source, and then decennial estimates are provided starting from 1800 and five-year estimates starting from 1950. Unless otherwise indicated, the figures for the period 1950-2050 come from the estimates (up to 2010) and projections (2010-2050) made by CELADE, and accessible online.

¹ The original figures for the period 1800–1950 were first evaluated graphically; then their year-on-year growth rates were studied, discarding any figures that led to inconsistent results. Once the baseline figures were established, the decennial estimates were calculated. The following notes specify the particularities of each case.

Mexico

Data on the total population of Mexico in the 19th century have been carefully compiled by Robert McCaa. Thanks to this, we have reliable decadal estimates between 1790 and 1910.

Modern censuses began in 1895, and census operations were carried out in 1900, 1910, 1921, 1930, 1940, 1950, 1960, 1970, 1980, 1990, 1995, 2000, 2005, and 2010. The only serious problem in

1 In: www.eclac.cl/celade/proyecciones/basesdatos_BD.htm, consulted in April and May 2012.

2 McCaa, Robert. "The Peopling of Nineteenth-Century Mexico: Critical Scrutiny of a Censored Century." In *Statistical Abstract of Latin America. Vol. 30, part 1*, edited by James W. et al. Wilkie, 603-33. Los Angeles: University of California at Los Angeles Latin American Center, 1993.; reviews in McCaa, Robert. "The People of Mexico from Origins to Revolution." In *A Population History of North America*, edited by Michael R. Haines and Richard H. Steckel, 241-304. Cambridge: Cambridge University Press, 2000.

The census data are the figures from 1921; the count, carried out at the end of the Mexican Revolution, had problems of coverage and under-enumeration, noted by various authors.³ Again, a recent work by Robert McCaa, based on retrospective projections of the 1930 census, allows us to solve the problem by proposing an acceptable correction of the 1921 figure.⁴

Guatemala

Population censuses were conducted in Guatemala in 1880, 1893, 1921, 1940, 1950, 1964, 1973, 1981, 1994, and 2002. The first four were officially corrected, with figures ranging from 3% in 1880 to 38% in 1940, based on poorly founded criteria. In this sense, the 1950 census can be considered the first based on modern statistical criteria.⁵ Estimates of the total population between 1880 and 1950 were made using the compensating equation (census figures and vital statistics) with different assumptions regarding corrections for underreporting and under-enumeration. Ultimately, the series that produced the most consistent results in terms of crude mortality and birth rates was chosen.⁶ Estimating the total population during the 19th century was more complicated. Before the first census in 1880, only very approximate data existed, based on estimates for electoral purposes, ecclesiastical calculations, etc. The main figures are shown in the table.

3 Collver, O. Andrew. *Birth rates in Latin America new estimates of historical trends and fluctuations*. Berkeley: Institute of International Studies, University of California, 1965., p. 141.

4 McCaa, Robert. "Missing millions: the demographic cost of the Mexican Revolution." *Mexican Studies/Estudios Mexicanos*. Vol. 19, No. 2 (2003), pp. 367-400.; see also Feliciano, Zaida M. «Mexico's Demographic Transformation: From 1920 to 1990.» In *A Population History of North America*, edited by Michael R. Haines and Richard H. Steckel, 601-30.

Cambridge: Cambridge University Press, 2000.; Mier y Terán, Marta. «Evolution of the Mexican population from the gifts of recenses: 1895-1970.» Doctoral thesis, Université de Montréal, 1982.

5 Collver, *Birth rates in Latin America new estimates of historical trends and fluctuations*., p. 124; Arias de Blois, Jorge. *The population of Guatemala*. Guatemala: CICRED - Central American Institute for Research and Industrial Technology 1976., pp. 9-15.

6 Pérez Brignoli, Héctor. *The population of Central America in the 20th century: annual vital series and estimates of the total population*. San José: Central American Population Center, 2000.

3 *The demographic transition in Latin America*

respective. Based on these data, Woodward 7 proposed some estimates for the period 1820-1870; in our case, we have adjusted Woodward's calculations to cover the period 1800-1880.

Honduras

The demographic data for Honduras are problematic. In the 1970s, after several surveys and assessments, demographers at CELADE determined that recorded deaths needed to be corrected by 70% to 100%, while births required adjustments of between 15% and 25%.⁸ Thus, it is understandable that, despite having more or less continuous vital statistics since 1910 and a series of population censuses, the data remains unreliable.

Despite the surveys conducted in 1881, 1887, 1901, 1905, 1910, 1916, 1926, 1930, 1935, 1940, 1945, 1950, 1961, 1974, 1988 and 2001, our knowledge of the history of the Honduran population is quite precarious.

Before 1950, the overall census figures show inconsistent results, with unreliable intercensal growth rates. A preliminary assessment of the data from 1910 onward, using census figures and vital statistics, led to the results shown in the corresponding table.⁹ The resulting series is relatively consistent and aligns well with CELADE estimates from 1970 onward; it involves correcting the 1910 enumerated population by 28%. This correction may seem very high, but it is the result of a back-projection of the 1930 enumerated population. The 19th-century figures are even more problematic. The original data available come from colonial registers, electoral and ecclesiastical estimates, and the first censuses conducted in 1881 and 1887; it was first necessary to adjust these figures to obtain a set of reasonable growth rates; then the decennial estimates were calculated.

7 Woodward, Ralph Lee. «Population growth in Central America during the first half of the century of national independence.» *Mesoamerica*. Vol. 1, No. 1 (1980), pp. 219-31.

8 See Pérez Brignoli, "The population of Central America in the 20th century: annual vital series and estimates of the total population." Section on Honduras.

9 Ibid.

El Salvador

Vital statistics are available from 1899 onwards, and national censuses were conducted in 1930, 1950, 1961, 1971, 1992, and 2007. The balancing equation produces consistent results for the period 1899-1994, once the underreporting of deaths is adjusted¹⁰ and an estimated population of 943,000 inhabitants in 1900 is calculated.¹¹ The population totals in the 19th century come from adjustments made to Woodward's calculations.

Nicaragua

Nicaragua has conducted national censuses in 1906, 1920, 1940, 1950, 1963, 1971, 1995, and 2005; the 1920 census is considered the first modern census. Vital statistics have been published since 1933, but underreporting of both deaths and births has always been very high; note, for example, that in the 1950s, estimates by CELADE considered underreporting of almost 130% for deaths and 25% for births.¹² Census figures, on the other hand, appear relatively consistent. For the 19th century, we have ecclesiastical data for 1815, electoral data for 1824, and then a series of estimates from a careful study by Alberto Lanuza.¹³

Costa Rica

The data come from my study on the population of Costa Rica, published in 2010¹⁴ and from official population projections.¹⁵

¹⁰ Ibid.

¹¹ Collver, *Birth rates in Latin America new estimates of historical trends and fluctuations.*, pp. 119-123.

¹² See Pérez Brignoli, "The population of Central America in the 20th century: annual vital series and estimates of the total population." Section on Nicaragua.

¹³ Lanuza Matamoros, Alberto. «Socioeconomic structures, power and state in Nicaragua, 1821-1875.» Undergraduate thesis, University of Costa Rica, 1976.

¹⁴ Pérez Brignoli, Héctor. *The population of Costa Rica, 1500-2000. An experimental history.* San José: Editorial Universidad de Costa Rica, 2010.

5 The demographic transition in Latin America

Panama

National censuses are available from 1911, 1920, 1930, 1940, 1950, 1960, 1970, 1980, 1990, 2000 and 2010, and the exhaustive study by Jaén Suárez for the 19th century.¹⁶ From 1911 to 2000, the figures include the population residing in the Canal Zone; from 2000 onwards, once the Republic of Panama assumed full control of the former Canal Zone, this territorial division disappears.

Cuba

The census data available for Cuba is quite extensive. In addition to the national censuses of 1907, 1919, 1931, 1953, 1970, 1981, and 2002, there are several enumerations and records compiled during the Spanish colonial period. The main tables from these censuses have been published retrospectively, which facilitates their access and use.¹⁷ From 1950 onward, we also have, as usual, the estimates and projections made by CELADE.

Haiti

The first modern census of Haiti was conducted in 1950. In 1945, the United States Census Bureau considered demographic information for Haiti to be deficient and only summary; vital statistics showed very high levels of underreporting, and censuses were lacking in the 19th and 20th centuries.¹⁸ The basic figures have been compiled from the studies of Engerman, Schaedel, Rotberg, and

¹⁵ Costa Rica, National Institute of Statistics and Censuses and University of Costa Rica, Central American Population Center. *Population estimates and projections by sex and age, 1950-2100*. San José: INEC-CCP, 2008.

¹⁶ Jaén Suárez, Omar. *The Population of the Isthmus of Panama from the 16th to the 20th Century. A Study of the Population and the Modes of Organization of the Economies, Societies, and Geographic Spaces*. Panama: Impresora de La Nación, 1979.

¹⁷ See Cuba, Institute of Statistical Research. *Population and housing censuses in Cuba. Estimates, censuses, and population counts of Cuba during the colonial period and the first American intervention*. Vol. 2, 2 vols. Havana: State Committee of Statistics, 1988. and Cuba, National Statistics Office. *Population and housing censuses in Cuba. Censuses of the period 1907-1953*. Havana: National Statistics Office, 2007.

¹⁸ United States, Bureau of the Census. *Haiti, summary of biostatistics. Maps and charts, population, natality and mortality statistics*. Washington, 1945.

Despeignes.¹⁹ For the period 1920-1940, Wilkie's estimates are retained. ²⁰ From 1950 onwards, the figures estimated by CELADE are used, produced from the censuses of 1950, 1971, 1982, and 2006 and several demographic surveys that have served to correct the still deficient vital statistics. The decline in the total population between 1791 and 1806, attributable to the slave rebellion, the expulsion of the white settlers, and the War of Independence, should be taken as a very rough estimate; the 1791 figure, due to Moreau de Saint-Méry, is accepted by authors such as Engerman and Knight²¹ while the 1806 figure comes from a census carried out during Dessaline's regime, cited by Schaedel.²²

Dominican Republic

The first modern census of the Dominican Republic took place only in 1950. Two earlier censuses, conducted in 1920 and 1935, yielded only partial data, published in an incomplete form. ²³ Estimates for the 19th century are particularly problematic.

For 1791 we have the calculations just cited by Moreau de St. Méry; the population decline, a product of wars and emigration, is noticeable afterwards and is reflected in the figures available for 1819 and 1825. Between 1822 and 1844 the Dominican Republic was under the

19 See Engerman, Stanley L. "A Population History of the Caribbean." In *A Population History of North America*, edited by Michael R. Haines and Richard H. Steckel, 483-528. New York: Cambridge University Press, 2000.; Schaedel, Richard P. *An essay on the human resources of Haiti*.

SI: USAID, Haiti, 1962.; Rotberg, Robert I. and Clague, Christopher K. *Haiti; the politics of squalor*. Boston: Houghton Mifflin, 1971. and Despeignes, Guy. «Politique de santé, medicalization of the society and emergence of modern clinical medicine in Haiti (1804-1915).» Doctoral thesis, Université de Montréal, 2001.

20 Wilkie, James W., Lorey, David E., and Ochoa, Enrique eds. *Statistical Abstract of Latin America*. Vol. 26. Los Angeles: UCLA. Latin American Center Publications. University of California, 1988., p. 77.

21 Moreau de Saint-Méry, MLE *Description topographique, physique, civile, politique et historique de la partie française de l'île Saint-Domingue*. Philadelphia, 1797.; Knight, Franklin W. *The Caribbean. The Genesis of a Fragmented Nationalism*. 2a. ed. New York: Oxford University Press, 1990., p. 367

22 Schaedel, *An essay on the human resources of Haiti*., p. 14.

23 See United States, Bureau of the Census. *Dominican Republic, summary of biostatistics. Maps and charts, population, natality and mortality statistics*. Washington, 1945.

7 The demographic transition in Latin America

Haitian rule. By 1850, the demographic recovery was evident and continued without major changes. But the figures are, in any case, approximate. A US estimate made in 1871 and cited by Hoetink²⁴ contradicts the figures from the ecclesiastical census of 1863. Only in the 20th century, and especially after the American occupation (1916-1924), did the data begin to improve. However, as already mentioned, we would have to wait until 1950 for the first truly modern census.

Puerto Rico

The first census by the U.S. administration was conducted in 1899; beginning in 1910, the population of Puerto Rico was counted every ten years as part of the United States census operation. The 1940 census publication reproduced the figures from the censuses carried out during the 19th century under Spanish administration.²⁵ From 1950 onwards, the estimates made by CELADE were used.²⁶

Venezuela

There are official estimates of Venezuela's total population for the entire 19th century;²⁷ however, these are inconsistent. The figures for the first half of the 19th century are based on very tentative estimates, which undoubtedly explains the numerous irregularities. The situation is different in the second half of that century and the first decades of the 20th. The census figures of 1873, 1881, 1891, 1920, and 1926 show a very slow growth rate, and even a negative one in the 1910s. Although Venezuela

²⁴ Hoetink, H. *The Dominican people, 1850-1900: notes for a historical sociology*. Baltimore: Johns Hopkins University Press, 1982.

²⁵ See Bartlett, Frederick P. and Howell, Brandon. *Puerto Rico and its population problem*. Santurce: Government of Puerto Rico. Board of Planning, Urbanization and Zoning, 1946.

²⁶ CELADE. «Latin America and the Caribbean: Population Estimates and Projections, 1950-2050." *Demographic Bulletin*. Vol. 73, No. 1-247 (2004), pp. In the online database, available in 2012, CELADE does not include relative estimates for Puerto Rico.

²⁷ Izard, Miguel. *Statistical series for the history of Venezuela*. Mérida: Universidad de los Andes, 1970., pp. 9-10.

Venezuela experienced serious political and economic upheavals during that period—one need only recall the blockade of 1902-03 and the external debt crisis—but this does not seem sufficient to explain a demographic trend like the one shown in official data. If this were true, the Venezuelan case would be entirely exceptional compared to the rest of the Latin American countries. Collver was the first to work on this problem and proposed a relatively acceptable adjustment, although it implies considering that the census figures of 1873, 1881, and 1891 over-enumerate the Venezuelan population.²⁸ Collver's correction is based on a projection of the Venezuelan population from 1885 to 1964, using vital statistics (with corrections, and estimates for the period 1895-1904 in which vital statistics are interrupted) and the age structure recorded in the censuses of 1936 and 1950. In our case, we start from Collver's estimates and perform calculations using the compensating equation,²⁹ from 1870 onwards; this led us to slightly increase the figures for the total population, thereby shortening the gaps with the total population derived from the censuses of 1873, 1881, and 1891; Everything indicates that the 1920 census significantly undercounted the total population of Venezuela. The results obtained appear consistent and agree with all the data from the period 1940-1965, for which the available information is abundant and reliable. From 1950 onward, the estimates made by CELADE, very similar to those obtained by Collver, are used. For the period prior to 1870, it was decided to retain the estimates based on the most careful studies, always bearing in mind that they do not result from modern censuses but from enumerations based on registers where the colonial administrative tradition was followed; thus, we have retained the estimates of Alexander von Humboldt, Agustín Codazzi, and the official estimates.

28 Collver, *Birth rates in Latin America new estimates of historical trends and fluctuations.*, pp. 164-173.

29 The compensating equation allows the population to be calculated annually, starting from an estimated figure on January 1 of the initial year, and then adding the births and subtracting the deaths that occurred throughout the year, or period considered; if the net migration balance is available for each period, this is added algebraically to the total population estimated for each year.

30 Codazzi, Agustín. *Summary of the geography of Venezuela*. Paris: H. Fournier and Company Printing House, 1841.; Humboldt, Alexander von. *Voyage to the equinoctial regions of the New Continent*.

9 The demographic transition in Latin America

Colombia

Census data for Colombia is abundant, in contrast to the scarcity and deficiency of vital statistics. The main tables of censuses prior to 1905 have been compiled in a volume of historical statistics, and we also have an extensive overview study by José Olinto Rueda.³¹ The figures from the 19th-century censuses appear quite consistent, except for the 1825 census. In this case, we have retained the adjusted figure proposed by Melo.³² In our calculations of Colombia's total population, we have excluded the Panamanian population, as Panama became an independent state in 1903.

Ecuador

Ecuadorian demography in the first half of the 19th century has been illuminated by Hammerly's studies, while for the second half of the century we have a perceptive article by Yves Saint-Geours.³³ From 1915 onwards we have Collver's estimates,³⁴ and from 1950 onwards, the calculations of CELADE.

Peru

Translated by Lisandro Alvarado; Eduardo Röhl and José Nucete Sardi. 5 vols. Caracas: Monte Ávila Editores, 1991. Originally published as 1941-42.; Izard, *Statistical Series for the History of Venezuela*.

31 See Urrutia, Miguel and Arrubla, Mario (Editors). *Compendium of Historical Statistics of Colombia*. Bogotá: National University of Colombia, 1970., pp. 9-30 and Rueda, José Olinto. "Demographic Processes in Colombia." In *One Hundred Years of Economy in Colombia*, edited by Rodrigo et al. Manrique, vol 2, 11-124. Bogotá: Externado. Journal of the Externado University of Colombia, 1986.

32 See Melo, Jorge Orlando. «The economic evolution of Colombia, 1830-1900». In *Manual de Historia de Colombia*, edited by Jaime Jaramillo Uribe, vol II, 135-210. Bogotá: Instituto Colombiano de Cultura, 1979., p. 138.

33 See Hamerly, Michael T. . . *Social and economic history of the former Province of Guayaquil, 1763-1842*. Guayaquil: Historical Archive of Guayas, 1973. and Saint-Geours, Yves. "L'évolution démographique de l'Équateur au XIXe siècle." In *Équateurs 1986*, edited by Daniel Delaunay and Michel Portais, vol 1, 197-207. Paris: Editions de l'ORSTOM, 1989.

34 Collver, *Birth rates in Latin America new estimates of historical trends and fluctuations.*, pp. 114-118.

There are several censuses for the end of the colonial period and the 19th century. These figures have been reviewed in a careful article by Paul Gootenberg, 35 which also reveals a previously unknown census conducted in 1827. The 1876 census, carried out shortly before the start of the War of the Pacific (1879–1884), is accepted by all scholars and constitutes an essential basis for evaluating earlier and later figures. However, it is not until 1940 that the first truly modern census of Peru is conducted. This leaves a long period of 64 years without census data. From 1950 onward, as is customary, we have the figures compiled by CELADE. We have obtained estimates of the total population for 1925, 1930 and 1940 by using the 1940 census and the vital statistics for the period 1925–1950 adjusted by Collver.³⁶ Given the impact of the War of the Pacific it was judged acceptable to estimate the population in 1890 by projecting the figure of 1876 with an average annual growth rate of 0.5%.

Between 1890 and 1925, intermediate estimates were obtained using the growth rate between the newly estimated population in 1890 and the figure calculated for 1925 from the population of 1940; this growth rate was 1.59% per year.

Bolivia

Figures for the total population of Bolivia are quite problematic before 1950, when the first modern census took place. The figure for 1792 is a very rough estimate, derived from the total number of tributary Indians recorded by Matías Bernal;³⁷ figures for the time of independence

35 See Gootenberg, Paul. "Population and Ethnicity in Early Republican Peru: Some Revisions." *Latin American Research Review*. Vol. 26, No. 3 (1991), pp. 109-57.

36 See Collver, *Birth rates in Latin America new estimates of historical trends and fluctuations.*, pp. 158-162.

37 Bernal recorded 123,174 tributaries, See Sánchez-Albornoz, Nicolás. *Indians and tributes in Upper Peru*. Lima: Institute of Peruvian Studies, 1978., p. 41; considering a multiplier of 4.5 to obtain the total indigenous population and assuming that the population of Upper Peru was composed of 75% indigenous people, the tentative figure of 740,000 is reached.

11 *The demographic transition in Latin America*

They range from Dalence's estimate,³⁸ of slightly less than one million, to the estimate of the English consul John Barclay Pentland of 1,100,000 and Brackenridge's exaggerated estimate of 1,700,000 made in 1818.³⁹ The calculation of 1847, made by Dalence from censuses carried out in 1845 and 1846, seems quite reliable; the author compiled the figures by departments and calculated at 760,000 the number of "infidel tribes" who lived outside the borders controlled by the republic, but he did not add these figures to the total registered population which resulted in 1,373,896 souls. Dalence also compiled information on births and deaths in 1826, 1831, 1836, 1841, and 1846, and with this information estimated the total population in 1825. The crude birth rate (45 per thousand) and death rate (32 per thousand) resulting from Dalence's data do not seem unreasonable and would imply a natural growth rate of 1.3% per year.⁴⁰ There were apparently census operations in 1854 and 1882, before the 1900 census, but we only have a few aggregate figures published in the 1900 census.⁴¹ The 1854 figure seems too high compared to that of 1846, as it would imply an annual growth rate of 2.36%; the 1882 figure records a smaller population than in 1846, which also does not seem realistic. Although Bolivia suffered significant territorial losses to Chile, Peru, and Brazil in the second half of the 19th century, these territories were relatively sparsely populated; nor can we attribute the decrease to the War of the Pacific, since, as is well known, the war was fought along the coast and did not directly affect the Altiplano, where the vast majority of the population lived.

38 Dalence, José María. *Statistical Sketch of Bolivia*. Chuquisaca (Sucre): Imprenta de Sucre, 1851., p. 217.

39 Pentland's figure appears in Fifer, J. Valerie. *Bolivia. Territory, situation and politics since 1825*. Translated by Sergio Aguirre Mac-Kay. Buenos Aires: Editorial Francisco de Aguirre, SA, 1976., p. 23, note 12; Brackenridge's calculation appears in Brackenridge, HM *Voyage to South America performed by order of the American Government in the years 1817 and 1818 in the Frigate Congress*. 2 vols. Baltimore, 1819., vol 2, p. 148. Both figures have been frequently reproduced in the geographical and statistical works of the 19th century.

40 See Dalence, *Statistical Sketch of Bolivia*, pp. 196-218.

41 See Bolivia, National Office of Immigration, Statistics and Geographical Propaganda. *General Census of the Population of the Republic of Bolivia according to the census of September 1, 1900*. 2 vols. La Paz: JM Gamarra, 1902-04.

Bolivian population. If we disregard the figures for 1854 and 1882 and consider those of 1846 and 1900 as more realistic, the Bolivian population grew slowly in the second half of the 19th century, at an annual rate of 0.46%. This rate increased during the 20th century, to 0.8% annually between 1900 and 1950, but this is quite low compared to the growth rates of other Latin American populations during the same period. For now, we must be content with this limited information.

In the case of Bolivia, we have retained two data series: one removing the apparently anomalous figures for 1854 and 1882, and the other retaining them. For the decennial estimates of the total population, we have considered only the series that removes the figures for 1854 and 1882.

Chili

The Chilean government conducted population censuses regularly from 1835 onward. Political stability and sound administrative organization ensured both the quality and early availability of Chilean statistics. No other Latin American country has such a frequent and regular series of censuses, and even fewer have annual vital statistics from 1850 onward. These advantages are complemented by the excellent statistical compilations carried out by Markos Mamalakis⁴² and a team of researchers from the Institute of Economics at the Catholic University of Chile.⁴³ Thus, estimates of the total population are taken from census data adjusted by the National Institute of Statistics and Censuses for the period 1800–1950.⁴⁴

From that date onwards, the estimates made by CELADE are recorded. The figures for 1800 and 1813 come from

⁴² Mamalakis, Markos. *Historical Statistics of Chile. Vol. 2. Demography and Labor Force*. Connecticut: Greenwood Press, 1980.

⁴³ Braun, Juan; Braun, Matías; Briones, Ignacio; Díaz, José. *Chilean Economy, 1810-1995. Historical Statistics*. Working Paper No. 187. Santiago, Chile: Pontifical Catholic University of Chile. Institute of Economics, 2000.

⁴⁴ We follow the data compiled in Gutiérrez Roldán, Héctor. *The population of Chile*. Paris: CICRED, 1975.

13 *The demographic transition in Latin America*

from estimates made from colonial censuses and are consistent with the figures available from 1835 onwards.

Brazil

Brazil has a remarkable series of national censuses, conducted in 1872, 1890, 1900, 1920, 1940, 1950, 1960, 1970, 1980, 1991, 2000, and 2010. Unfortunately, vital statistics at the national level are only available from 1959 onward. There is also no highly reliable data on the total population before the first national census, conducted in 1872. Giorgio Mortara⁴⁵ compiled and studied the available figures for the late 18th and throughout the 19th centuries and produced a coherent series of estimates, which has been used as the basis for all demographic studies⁴⁶ and also for the official estimates made by the Brazilian Institute of Geography and Statistics (IBGE).⁴⁷

Paraguay

The figure of 1796 corresponds to the estimates of Félix de Azara⁴⁸ and is accepted as reasonable by all scholars of Paraguay. The 1846 figure comes from a census carried out by order of Bishop Basilio López and adjusted by Domingo Rivarola.⁴⁹ The War of the Triple Alliance or Paraguayan War (1864-1870) undoubtedly decimated the

45 Mortara, Giorgio. «Studies on the use of the demographic census for the reconstitution of statistics of the population movement of Brazil.» *Brazilian Journal of Statistics*. Vol. III, No. 5 (1941), pp.

46 Merrick, Thomas William and Graham, Douglas H. *Population and economic development in Brazil, 1800 to the present*. Baltimore: Johns Hopkins University Press, 1979.; Berquo, Elza et al. *The population of Brazil*. Paris: CICRED, 1975.

47 Brazil, IBGE. *Contributions to the demographic study of Brazil*. Rio de Janeiro: IBGE - National Statistical Council, 1961.; Brazil, IBGE. *Historical Statistics of Brazil. Economic, Demographic and Social Series from 1550 to 1985*. Rio de Janeiro: IBGE, 1987.

48 Azara, Félix de, Azara, Agustín de, and Castellanos de Losada, Basilio Sebastián. *Description and history of Paraguay and the Rio de la Plata*. 2 vols. Madrid: Impr. de Sanchiz, 1847., volume 1, p. 330.

49 See Rivarola, Domingo M. *La Población del Paraguay*. Asunción: Centro Paraguayo de Estudios Sociológicos, 1974., p. 11. See also Williams, John Hoyt. «Observations on the Paraguayan Census of 1846.» *Hispanic American Historical Review*. Vol. 56, No. 3 (1976), pp. 424-37.

Paraguayan population, but it is not easy to establish reliable figures. Although debatable, Vera Blinn Reber's study is the most comprehensive available to date (2011).⁵⁰ Figures relating to the total population at the start of the war range from 312,000 to over one million inhabitants, although most of these figures are based on rough, poorly substantiated calculations. Vera Blinn Reber estimates a total of 318,114 inhabitants from a household register by district, carried out by the government in 1864, assuming an average household size of 6 people; similar calculations are possible for 1865, 1866, and 1867, years in which household registers by district were also carried out. Increasing the average household size to 7, the total population would increase to about 353,000 inhabitants. Another method of

The calculation used, among others by Gabriel Carrasco, is an estimate based on the size of the army mobilized by Paraguay at the beginning of the war. In this case, the difficulty lies in accurately establishing what proportion of the male population aged 15 to 50 was incorporated into the army. Using different hypotheses, Carrasco estimates a minimum population in 1865 of 320,000 inhabitants and a maximum of 525,000. In short, the total population of Paraguay around 1865 appears to have been around the 350,000 inhabitants; Vera Blinn Reber's figures are therefore acceptable. Establishing a figure for the total population after the war is over is even more complex. Carrasco accepts Schüttere's estimate of 231,000 inhabitants for 1872; the Dutch geographer Jan Kleipenning indicates a similar figure, 221,079 inhabitants, based on German consular sources; Potthast and Whigham, using the registers of an unpublished but partial census recently discovered in the archives of Asunción, estimate the population of 1872 at 116,351 inhabitants. Comparing the estimates of Carrasco, Reber, Whigham-Potthast, and Kleipenning, the percentage of

50 Reber, Vera Blinn. "The Demographics of Paraguay: A Reinterpretation of the Great War, 1864-70." *Hispanic American Historical Review*. Vol. 68, No. 2 (1988), pp. 290-319..

51 Carrasco, Gabriel. *The population of Paraguay, before and after the war; rectification of generally accepted opinions*. Asunción: National Workshops of H. Kraus, 1905., p. 4. Gabriel Carrasco was a geographer, statistician and politician from Santa Fe.

52 Whigham, Thomas L. and Potthast, Barbara. "The Paraguayan Rosetta Stone: New Insights into the Demographics of the Paraguayan War, 1864-1870." *Latin American Research Review*. Vol. 34, No. 1 (1999), pp. 174-86..

15 *The demographic transition in Latin America*

The decline in the Paraguayan population between the beginning and end of the war would thus range between 17% and 75%. Obviously, the highest estimate comes from the 116,351 inhabitants estimated by Whigham and Potthast for 1872. How acceptable is this figure? An assessment can be made by considering the evolution of the Paraguayan population up to the end of the 19th century; by this time, the total population was approaching half a million inhabitants, a figure consistent with the demographic history of the 20th century and accepted by almost all authors. The implied average annual growth rates,

Comparing the various estimates for 1872 with the accepted endpoint in 1899, they range from 1.9% to 5.3%. Given Paraguay's demographic conditions, it is very difficult to accept annual growth rates much higher than 2%. To illustrate, a 2% annual rate would imply, for example, a birth rate of 40 per thousand and a death rate of 20 per thousand. It is difficult to imagine, given the conditions of the time, a higher birth rate and, above all, a lower death rate than the indicated limit. It goes without saying that an annual growth rate of 5.3%, such as that resulting from Whigham and Potthast's figures, is simply impossible.⁵³ The problem does not stem from the 1872 census data, discovered and analyzed by both authors, but from the fact that they consider it complete, that is, free of underreporting.

The Paraguayan censuses of 1846 and 1886 suffered from this problem, and it seems the same occurred with the 1872 census. Note that the 1886 census was officially corrected, first by 10% and then by 37.5%. In situations like this, we are operating in an area where figures are highly unreliable, and we can only use, to validate them,

consistency criteria. The examination of implicit growth rates is one of these criteria, and it was a great merit of Vera Blinn Reber to apply it for the first time to Paraguayan figures from the 19th century in her 1988 article.

⁵³ A similar argument is developed in Kleinpenning, Jan MG "New Insights into the Demographics of the Paraguayan War." *Latin American Research Review*. Vol. 37, No. 3 (2002), pp. 137-42.. See also Whigham, Thomas L. and Potthast, Barbara. "Refining the Numbers: A Response to Reber and Kleinpenning." *Latin American Research Review*. Vol. 37, No. 3 (2002), pp. 143-48..

The first modern census of Paraguay was carried out in 1950, and from that date onwards we have the estimates made by CELADE. Between 1899 and 1950 we have retained the figures on the total population calculated by Raúl Mendoza and accepted, among others, by Rivarola and Kleinpenning.⁵⁴

Argentina

The available figures on the total population of Argentina from 1800 onwards are consistent and reliable. In addition to the national censuses of 1869, 1895, 1914, 1947, 1960, 1970, 1980, 1991, 2001, and 2010, other statistical data are included, and above all, a careful evaluation of the available figures carried out by Alfredo Lattes;⁵⁵ two series are included, one with estimates of the unreduced indigenous population between 1800 and 1870 and another excluding this population group.⁵⁶

From 1950 onwards, the figures estimated by CELADE are followed.

Uruguay

In the case of Uruguay, national censuses are available for 1852, 1860, 1908, 1963, 1975, 1986, 1996, 2004, and 2011; as can be seen, there is a significant gap in the first half of the 20th century, and very scattered data for the 19th century. In fact, the very constitution of Uruguay as a nation-state during the first decades of the 19th century is problematic. As baseline figures, we have chosen the estimates of Adela Pellegrino,⁵⁷ adding data from Oddone⁵⁸ and

54 See Kleinpenning, Jan MG *Rural Paraguay, 1870-1963. A Geography of Progress, Plunder and Poverty*. Vol. 1. 2 vols. Madrid / Frankfurt: Iberoamericana - Vervuert, 2009.; Mendoza, Raúl. *Population trends and demographic problems of Paraguay*. Asunción: Paraguayan Center for Population Studies, 1968.; Rivarola, *The Population of Paraguay*.

55 Recchini de Lattes, Zulma L. and Lattes, Alfredo E. *The Population of Argentina*. Demographic Research Series. Buenos Aires: Republic of Argentina, Ministry of Economy, State Secretariat of Economic Programming and Coordination, National Institute of Statistics and Censuses, 1975.

56 The figures on the total population excluding the unreduced indigenous people come from Maeder, Ernesto JA *Evolución demográfica argentina de 1810 a 1869*. Buenos Aires: Editorial Universitaria de Buenos Aires, 1969.

57 Pellegrino, Adela. *Demographic Characterization of Uruguay*. Montevideo: Population Program, Faculty of Social Sciences, University of the Republic, 2003.; see also Rial Roade, Juan. *Historical Statistics of Uruguay, 1850-1930: population, agricultural production, trade,*

17 *The demographic transition in Latin America*

Nahum,⁵⁹ and also the census figures. The resulting series is consistent but it should be borne in mind that it is only tentative for much of the 19th century. From 1950 onwards, we use the CELADE estimates.

Literature

Arias de Blois, Jorge. *The population of Guatemala*. Guatemala: CICRED - Central American Institute for Research and Industrial Technology 1976.

Azara, Félix de, Azara, Agustín de, and Castellanos de Losada, Basilio Sebastián. *Description and history of Paraguay and the Río de la Plata*. 2 vols. Madrid: Impr. by Sanchiz, 1847.

Bartlett, Frederick P. and Howell, Brandon. *Puerto Rico and its population problem*. Santurce: Government of Puerto Rico. Planning, Urbanization and Zoning Board, 1946.

Berquo, Elza et al. *The population of Brazil*. Paris: CICRED, 1975.

Bolivia, National Office of Immigration, Statistics and Geographical Propaganda. *General census of the population of the Republic of Bolivia according to the census of September 1, 1900*. 2 vols. La Paz: JM Gamarra, 1902-04.

Brackendridge, HM *Voyage to South America performed by order of the American Government in the years 1817 and 1818 in the Frigate Congress*. 2 vols. Baltimore, 1819.

Brazil, IBGE. *Contributions to the demographic study of Brazil*. Rio de Janeiro: IBGE - National Statistical Council, 1961.

Industry, urbanization, communications, quality of life. Notebook. Montevideo: Center for Information and Studies of Uruguay, 1980.; Rial Roade, Juan. *Population and development of a small country Uruguay, 1830-1930*. Montevideo: Center for Information and Studies of Uruguay. Acali Editorial, 1983.

58 Oddone, Juan Antonio. *The Formation of Modern Uruguay. Immigration and Economic and Social Development*. Buenos Aires: Editorial Universitaria de Buenos Aires, 1966.

59 Nahum, Benjamín. *Manual of the History of Uruguay. Volume 1: 1830-1903*. Montevideo: Ediciones de la Banda Oriental, 1993.

Brazil, IBGE. *Historical Statistics of Brazil. Economic, Demographic and Social Series from 1550 to 1985*. Rio de Janeiro: IBGE, 1987.

Braun, Juan; Braun, Matías; Briones, Ignacio; Díaz, José. *Chilean Economy, 1810-1995. Historical Statistics*. Working Paper No. 187.
Santiago, Chile: Pontifical Catholic University of Chile. Institute of Economics, 2000.

Carrasco, Gabriel. *The population of Paraguay, before and after the war; rectification of generally accepted opinions*. Asunción: H. Kraus National Workshops, 1905.

CELADE. «Latin America and the Caribbean: Population Estimates and Projections, 1950-2050». *Demographic Bulletin*. Vol. 73, No. 1-247 (2004), pp.

Codazzi, Agustín. *Summary of the geography of Venezuela*. Paris: H. Printing House. Fournier and Company, 1841.

Collver, O. Andrew. *Birth rates in Latin America new estimates of historical trends and fluctuations*. Berkeley: Institute of International Studies, University of California, 1965.

Costa Rica, National Institute of Statistics and Censuses and University of Costa Rica, Central American Population Center. *Population estimates and projections by sex and age, 1950-2100*. San José: INEC-CCP, 2008.

Cuba, Institute of Statistical Research. *Population and Housing Censuses in Cuba. Estimates, Registrations and Population Censuses of Cuba from the Colonial Era and the First American Intervention*. Vol. 2, 2 vols. Havana: State Committee of Statistics, 1988.

Cuba, National Statistics Office. *Population and Housing Censuses in Cuba. Censuses of the period 1907-1953*. Havana: National Statistics Office, 2007.

Dalence, José María. *Statistical Sketch of Bolivia*. Chuquisaca (Sucre): Sucre Printing Press, 1851.

19 *The demographic transition in Latin America*

Despeignes, Guy. "Policy of health, medicalization of the society and emergence of modern clinical medicine in Haiti (1804-1915)."
Doctoral thesis, Université de Montréal, 2001.

Engerman, Stanley L. "A Population History of the Caribbean." In *A Population History of North America*, edited by Michael R. Haines and Richard H. Steckel, 483-528.
New York: Cambridge University Press, 2000.

Feliciano, Zaida M. "Mexico's Demographic Transformation: From 1920 to 1990." In *A Population History of North America*, edited by Michael R. Haines and Richard H. Steckel, 601-30. Cambridge: Cambridge University Press, 2000.

Fifer, J. Valerie. *Bolivia. Territory, situation and politics since 1825*. Translated by Sergio Aguirre Mac-Kay. Buenos Aires: Editorial Francisco de Aguirre, SA, 1976.

Gootenberg, Paul. "Population and Ethnicity in Early Republican Peru: Some Revisions."
Latin American Research Review. Vol. 26, No. 3 (1991), pp. 109-57.

Gutiérrez Roldán, Héctor. *The population of Chile*. Paris: CICRED, 1975.

Hamerly, Michael T. . *Social and economic history of the former Province of Guayaquil, 1763-1842*. Guayaquil: Historical Archive of Guayas, 1973.

Hoetink, H. *The Dominican people, 1850-1900: notes for a historical sociology*.
Baltimore: Johns Hopkins University Press, 1982.

Humboldt, Alexander von. *Voyage to the equinoctial regions of the New Continent*.
Translated by Lisandro Alvarado; Eduardo Röhl and José Nucete Sardi. 5 vols.
Caracas: Monte Ávila Editores, 1991. Originally published as 1941-42.

Izard, Miguel. *Statistical series for the history of Venezuela*. Mérida: University of the Andes, 1970.

Jaén Suárez, Omar. *The population of the Isthmus of Panama from the 16th to the 20th century. Study on population and modes of economic organization, the*

Societies and geographical spaces. Panama: La Nación Printers, 1979.

Kleinpenning, Jan MG "New Insights into the Demographics of the Paraguayan War."
Latin American Research Review. Vol. 37, No. 3 (2002), pp. 137-42.

Kleinpenning, Jan MG *Rural Paraguay, 1870-1963. A Geography of Progress, Plunder and Poverty*. Vol. 1. 2 vols. Madrid / Frankfurt: Iberoamericana - Vervuert, 2009.

Knight, Franklin W. *The Caribbean. The Genesis of a Fragmented Nationalism*. 2a. ed.
New York: Oxford University Press, 1990.

Lanuza Matamoros, Alberto. «Socioeconomic structures, power and state in Nicaragua, 1821-1875.» Undergraduate thesis, University of Costa Rica, 1976.

Maeder, Ernesto JA *Argentine demographic evolution from 1810 to 1869*. Buenos Aires
Aires: Editorial Universitaria de Buenos Aires, 1969.

Mamalakis, Markos. *Historical Statistics of Chile*. Vol. 2. *Demography and Labor Force*.
Connecticut: Greenwood Press, 1980.

McCaa, Robert. "The Peopling of Nineteenth-Century Mexico: Critical Scrutiny of a
Censored Century." In *Statistical Abstract of Latin America*. Vol. 30, part 1,
edited by James W. et al. Wilkie, 603-33.
Los Angeles: University of California at Los Angeles Latin American Center,
1993.

McCaa, Robert. "The People of Mexico from Origins to Revolution." In *A Population
History of North America*, edited by Michael R. Haines and Richard H. Steckel,
241-304. Cambridge: Cambridge University Press, 2000.

McCaa, Robert. "Missing millions: the demographic cost of the Mexican Revolution."
Mexican Studies/Estudios Mexicanos. Vol. 19, No. 2 (2003), pp. 367-400.

Melo, Jorge Orlando. "The economic evolution of Colombia, 1830-1900."
In *Manual de Historia de Colombia*, edited by Jaime Jaramillo Uribe, vol II,
135-210. Bogotá: Instituto Colombiano de Cultura, 1979.

21 *The demographic transition in Latin America*

Mendoza, Raúl. *Population trends and demographic problems in Paraguay*.

Asunción: Paraguayan Center for Population Studies, 1968.

Merrick, Thomas William and Graham, Douglas H. *Population and economic development in Brazil, 1800 to the present*. Baltimore: Johns Hopkins University Press, 1979.

Mier y Terán, Marta. «Evolution of the Mexican population from the gifts of recenses: 1895-1970.» Doctoral thesis, Université de Montréal, 1982.

Moreau de Saint-Méry, MLE *Description topographique, physique, civile, politique et historique de la partie française de l'isle Saint-Domingue*. Philadelphia, 1797.

Mortara, Giorgio. «Studies on the use of the demographic census for the reconstitution of statistics of the population movement of Brazil.» *Brazilian Journal of Statistics*. Vol. III, No. 5 (1941), pp.

Nahum, Benjamín. *Manual of the History of Uruguay. Volume 1: 1830-1903*. Montevideo: Ediciones de la Banda Oriental, 1993.

Oddone, Juan Antonio. *The Formation of Modern Uruguay. Immigration and Economic and Social Development*. Buenos Aires: Editorial Universitaria de Buenos Aires, 1966.

Pellegrino, Adela. *Demographic characterization of Uruguay*. Montevideo: Population Program, Faculty of Social Sciences, University of the Republic, 2003.

Pérez Brignoli, Héctor. *The population of Central America in the 20th century: annual vital series and total population estimates*. San José: Central American Population Center, 2000.

Pérez Brignoli, Héctor. *The Population of Costa Rica, 1500-2000. An Experimental History*. San José: Editorial Universidad de Costa Rica, 2010.

Reber, Vera Blinn. "The Demographics of Paraguay: A Reinterpretation of the Great War, 1864-70." *Hispanic American Historical Review*. Vol. 68, No. 2 (1988), pp. 290-319.

- Recchini de Lattes, Zulma L. and Lattes, Alfredo E. *The Population of Argentina*. Demographic Research Series. Buenos Aires: Republic of Argentina, Ministry of Economy, State Secretariat of Economic Planning and Coordination, National Institute of Statistics and Censuses, 1975.
- Rial Roade, Juan. *Historical Statistics of Uruguay, 1850-1930: Population, Agricultural Production, Commerce, Industry, Urbanization, Communications, Quality of Life*. Notebook. Montevideo: Center for Information and Studies of Uruguay, 1980.
- Rial Roade, Juan. *Population and development of a small country Uruguay, 1830-1930*. Montevideo: Uruguay Information and Studies Center. Acali Editorial, 1983.
- Rivarola, Domingo M. *The Population of Paraguay*. Asunción: Paraguayan Center for Sociological Studies, 1974.
- Rotberg, Robert I. and Clague, Christopher K. *Haiti; the politics of squalor*. Boston: Houghton Mifflin, 1971.
- Rueda, José Olinto. «Demographic Processes in Colombia». In *One Hundred Years of Economics in Colombia*, edited by Rodrigo et al. Manrique, vol 2, 11-124. Bogotá: Externado. Journal of the Externado University of Colombia, 1986.
- Saint-Geours, Yves. "L'évolution démographique de l'Équateur au XIXe siècle." In *Équateurs 1986*, edited by Daniel Delaunay and Michel Portais, vol 1, 197-207. Paris: Editions de l'ORSTOM, 1989.
- Sánchez-Albornoz, Nicolás. *Indians and tributes in Upper Peru*. Lima: Institute of Peruvian Studies, 1978.
- Schaedel, Richard P. *An essay on the human resources of Haiti*. SI: USAID, Haiti, 1962.
- United States, Bureau of the Census. *Dominican Republic, summary of biostatistics. Maps and charts, population, natality and mortality statistics*. Washington, 1945.
- United States, Bureau of the Census. *Haiti, summary of biostatistics. Maps and charts, population, natality and mortality statistics*. Washington, 1945.

23 *The demographic transition in Latin America*

Urrutia, Miguel and Arrubla, Mario (Editors). *Compendium of Historical Statistics of Colombia*. Bogotá: National University of Colombia, 1970.

Whigham, Thomas L. and Potthast, Barbara. «The Paraguayan Rosetta Stone: New Insights into the Demographics of the Paraguayan War, 1864-1870." *Latin American Research Review*. Vol. 34, No. 1 (1999), pp. 174-86.

Whigham, Thomas L. and Potthast, Barbara. "Refining the Numbers: A Response to Reber and Kleinpenning." *Latin American Research Review*. Vol. 37, No. 3 (2002), pp. 143-48.

Wilkie, James W., Lorey, David E., and Ochoa, Enrique eds. *Statistical Abstract of Latin America*. Vol. 26. Los Angeles: UCLA. Latin American Center Publications. University of California, 1988.

Williams, John Hoyt. "Observations on the Paraguayan Census of 1846." *Hispanic American Historical Review*. Vol. 56, No. 3 (1976), pp. 424-37.

Woodward, Ralph Lee. "Population Growth in Central America During the First Half of the Century of National Independence." *Mesoamerica*. Vol. 1, No. 1 (1980), pp. 219-31.